**University Of Engineering and Technology, Lahore**
**Computer Engineering Department**

# Lab Evaluation:

| | |
|---|---|
| **Course Name:** Data Structures & Algorithm | **Course Code:** CMPE-212L |
| **Assignment Type:** Lab | **Dated:** 30th Jan 2024 |
| **Semester:** 4th | **Session:** 2022 |
| **Lab/Project/Assignment #:** 3 | **CLOs to be covered:** CLO 1 |
| **Lab Title:** Pointers & Dynamic Memory Allocation | **Teacher Name:** Engr. Afeef Obaid |

| **CLO 1** | Evaluate known data structures and algorithms for problem solving | | | | | |
|---|---|---|---|---|---|---|
| **Levels (Marks)** | **Level 1** | **Level 2** | **Level 3** | **Level 4** | **Level 5** | **Level 6** |
| (10) | | | | | | |
| | | | | | **Total** | **/10** |

## Rubrics for Current Lab Evaluation

| Scale | Marks | Level | Rubric |
|---|---|---|---|
| Excellent | **9-10** | L1 | Submitted all lab tasks, BONUS task, have good understanding. |
| Very Good | **7-8** | L2 | Submitted the lab tasks but have good understanding |
| Good | **5-6** | L3 | Submitted the lab tasks but have weak understanding. |
| Basic | **3-4** | L4 | Submitted the lab tasks but have no understanding. |
| Barely Acceptable | **1-2** | L5 | Submitted only one lab task. |
| Not Acceptable | **0** | L6 | Did not attempt |

# LAB No. 3

**Lab Goals/Objectives:**

- Pointers and references
- Pointers and arrays
- Dynamic memory allocation.

**Equipment Required:** Computer and Dev C++ Compiler

**Theory/Relevant Material:** Pointers & Dynamic Memory Allocation

**Lab Tasks:**

1. Declare an array pointer. Ask user for initial size and allocate memory using ***new.***
2. Write separate function for insert, resize, remove and display.
3. When array utilization is around 70%, allocate a new array with double size, copy elements of original array into it, delete original array. Then set pointer pointing to original array to this new array.